Sahana Naganandh

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Education

University of Edinburgh — Edinburgh, United Kingdom
BEng Electronics & Computer Science (Year 3 Transfer), in progress, expected 2028 — *2026 – 2028*

Vellore Institute of Technology — Vellore, India
B.Tech Computer Science and Engineering (AI & ML), transferred after Year 2, CGPA 8.75/10 — *Transferred 2026*

Experience

AI Intern — Jun 2026 – Aug 2026
Dell Technologies — *Dubai, United Arab Emirates*

- Designed and built Python/FastAPI backend services and REST APIs for an AI-driven calling and recruitment assistant that automates candidate screening, supporting a 500-engineer service team and translating recruitment requirements into backend design with engineers
- Built an ElevenLabs-based voice calling agent that collects information directly from engineers and feeds it into the screening pipeline, integrating the voice API with the backend services and n8n-orchestrated workflows
- Shipped a resume parser now used by staff in the live screening workflow (NLP-based extraction), and architected n8n workflows orchestrating AI agents, external APIs, and multi-step automation

UI Intern — May 2025 – Jun 2025
Nectar Building Technologies Inc. — *Coimbatore, India*

- Worked on the Front-End Development team contributing to Nectar Connect, the company's core building-operations platform that connects, optimizes, and operates over 1,500 buildings globally

Research Assistant Intern — Jun 2023 – Aug 2023
Manipal Institute of Higher Education — *Dubai, United Arab Emirates*

- Prepared large datasets for AI/ML research by cleaning, structuring, and preprocessing them in Python (TensorFlow, NumPy) ahead of downstream analysis
- Built Power BI visualisations to identify trends and communicate data-driven insights to the research team

Projects & Publications

Portfolio & Trade Management Platform | *Java, Spring Boot, Python, FastAPI, PostgreSQL*

- Built a two-service platform pairing a Java Spring Boot service, which owns portfolio, order, and trade domain logic through REST APIs with PostgreSQL persistence, with a Python FastAPI service for historical price processing and risk analytics
- Implemented the end-to-end trading flow: multi-security price ingestion, buy/sell order validation and execution, cash and holdings updates, and realised/unrealised P&L, all persisted in PostgreSQL
- Designed a quantitative risk layer covering portfolio volatility, covariance-based diversification, VaR, and Expected Shortfall

AI Job Search Framework | *Claude Code, Python*

- Directed Claude Code to orchestrate a drafter-reviewer multi-agent workflow that evaluates job fit and drafts tailored CVs and cover letters, with a second agent critiquing each draft against a factual-grounding audit before anything is submitted
- Built a compile-and-inspect verification loop with automated ATS text-layer checks, so every generated document is validated for page length, layout, and parseability

OS-Level Optimization for Energy-Efficient AI Inference | *ICNDA 2026, Accepted (lead author)*

- Raised AI inference throughput by up to 50% and cut latency by up to 35% on ResNet-50 (YOLOv8: +33% FPS, -28% latency) against an untuned Linux baseline, by combining CPU governor tuning, GPU persistent mode and clock locking, NUMA-aware memory binding with numactl, CPU affinity pinning, and batch-size/thread tuning
- Benchmarked both models in PyTorch/CUDA on Ubuntu 22.04 across 5 sequential tuning stages, running each configuration 5 times with under 2% variability and 1.5x IQR outlier removal, measuring single-stream latency and offline throughput
- Logged latency, FPS, utilisation, and memory bandwidth per run with a custom Python script (nvidia-smi, htop, perf)
- Isolated each tuning stage in sequence so every gain could be attributed to a single change, with software versions locked and thermals monitored to keep runs comparable

AI and Privacy Concerns: Balancing Innovation With Security | *Cureus, 2025, Published* — Visit paper

- Assessed PII exposure across 4 languages (English, French, German, Italian) in multilingual AI datasets with named entity recognition and token-count analysis, evaluating PII43K masking and redaction as a GDPR/CCPA-compliant alternative
- Synthesised privacy-preserving techniques, including differential privacy, federated learning, secure multi-party computation, and explainable AI, into regulatory-compliance recommendations for privacy-aware AI model design

Technical Skills

Languages: Python, Java, C/C++, SQL
Backend & Data: Spring Boot, FastAPI, REST APIs, PostgreSQL, n8n
Systems & Tools: Linux, Git, Postman, Claude Code (AI-assisted development)
Data & ML: NumPy, Pandas, PyTorch, TensorFlow, Power BI
Fundamentals: Data Structures & Algorithms, Object-Oriented Programming

Leadership & Achievements

1st Place, Murdoch University Hackathon, Dubai – *built a volunteering-services app in 3 days with a 5-person team*

Winner, GITEX Teens in AI Hackathon

Finalist, Top 10 of 2,000+ entrants, Google DSC DevJams Hackathon – *built Klyra, an AI-assisted learning app with automated quiz, summary, and flashcard generation, Wikipedia/YouTube API integration, and backend content-filtering and question-generation logic*

Senior Core Member, IEEE Computer Society, VIT – *developed a terminal-based, puzzle-driven C++ executable and contributed to technical ideation and execution planning for a large-scale student tech event*

Senior Core Member, CodeChef, VIT – *contributed to MERN stack projects and supported event coordination for high-participation competitive coding events*